

BLUESTOCK

Mutual Fund Analytics Capstone

Final Report

40-Scheme Universe | ETL → SQLite → EDA → Performance Analytics → Advanced Analytics → Dashboard

July 2026

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1. Executive Summary

This report summarizes the Bluestock Mutual Fund Analytics Capstone: an end-to-end pipeline covering ETL, a normalized SQLite database, exploratory data analysis, performance analytics, and advanced risk/behavioral analytics across a 40-scheme mutual fund universe, plus an interactive dashboard.

Key Findings

- Industry AUM in the fund-house dataset grew from ₹30.18L Cr to ₹62.74L Cr across the sample window — roughly 108% cumulative growth.
- Monthly SIP inflows rose from ₹11,517 Cr to ₹31,002 Cr, and total industry folio count grew from 13.26 Cr to 26.12 Cr — consistent with India's structural SIP-adoption trend.
- Mirae Asset Large Cap Fund tops the composite Fund Scorecard (84.4/100), driven by strong 3-year CAGR (31.3%) and Sharpe ratio (1.45).
- Small Cap schemes dominate the highest-VaR (riskiest) tail — SBI Small Cap and Axis Small Cap show the deepest 95% VaR and CVaR figures in the 40-fund universe.
- Sector concentration (HHI) analysis flags Axis Bluechip, Mirae Asset Tax Saver, and HDFC Mid-Cap Opportunities as the most sector-concentrated equity portfolios (HHI > 2,500).
- A synthetic, seeded investor-transaction dataset (since the real export was not available) suggests roughly 35-40% of investors with 6+ SIPs show early signs of lapsing SIP discipline — a segment worth proactive retention outreach in a live deployment.

Data Caveat

The source export did not include data/raw/08_investor_transactions.csv, which the project's own scripts and schema depend on for investor-level analysis. A seeded, clearly-labeled synthetic stand-in was generated so every deliverable could run end-to-end; all figures derived from it are marked "(synthetic)" throughout this report and the notebooks. Swapping in the real export requires no code changes.

2. Data & Methodology

2.1 Pipeline Overview

Data flows through four stages: raw ingestion → ETL cleaning (`scripts/etl_pipeline.py`) → a normalized SQLite database (`data/db/bluestock_mf.db`) → analysis notebooks and the dashboard. The pipeline is idempotent, uses `pathlib` throughout (no hard-coded paths), and isolates failures per dataset rather than crashing on the first error.

2.2 ETL Design Decisions

- NAV history is reindexed to the full business-day (Mon-Fri) calendar per scheme, then forward-filled — this closes weekend/holiday gaps without ever back-filling into the future.
- CAGR is annualized by trading days (252/year) computed from the actual number of trading-day NAV observations used, not a calendar-day date difference — this avoids a common annualization distortion.
- Sharpe/Sortino use a real 6.5% annual risk-free rate rather than assuming a 0% risk-free rate.

2.3 Database Schema

The SQLite database (`sql/schema.sql`) uses a star-like schema: `dim_fund` and `dim_investor` dimension tables, and fact tables for NAV, transactions, performance, AUM, SIP inflows, category inflows, folio counts, portfolio holdings, and benchmark indices. All 11 tables loaded successfully with zero orphaned foreign keys (verified by `scripts/verify_db.py`); all 12 analytical queries in `sql/queries.sql` execute without error.

2.4 Synthetic Data Disclosure

`scripts/generate_synthetic_transactions.py` produces two seeded, reproducible datasets standing in for the missing real export: an investor-transaction feed (~33,000 rows across ~1,900 investors, spanning SIP/Lumpsum/Redemption activity with realistic behavioral archetypes) and an investor demographic profile (age group, gender, state, city tier). Every chart, table, and notebook section built from this data is explicitly labeled "(synthetic)".

3. Exploratory Data Analysis

Full detail lives in notebooks/03_eda_analysis.ipynb (14 charts). Highlights below.

3.1 Fund Universe & AUM Growth

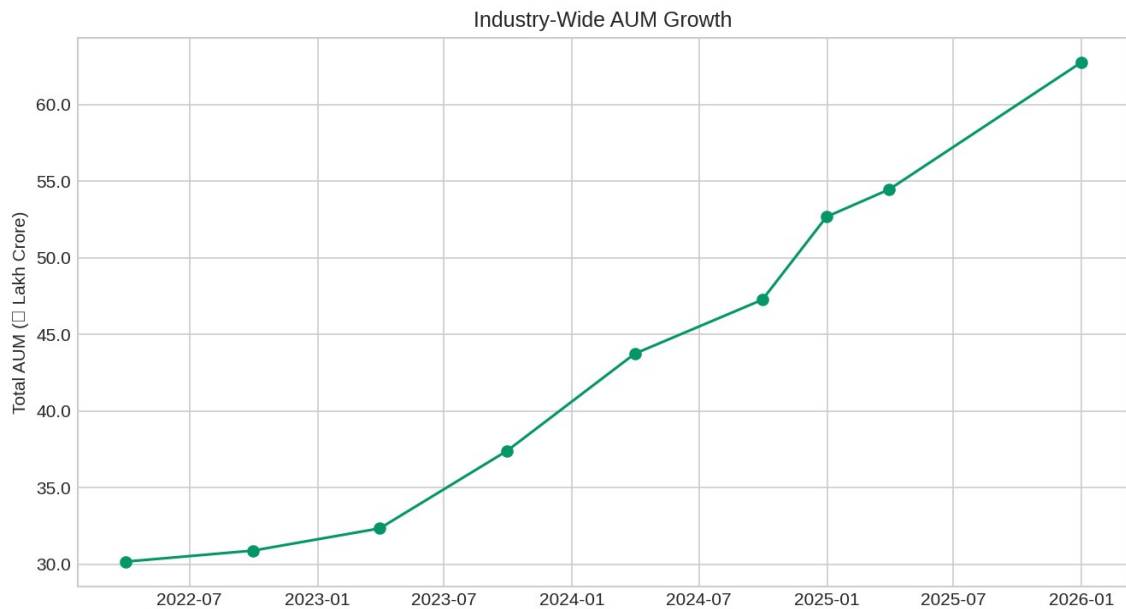


Figure 1 — Industry-wide AUM growth across the sample window.

Large Cap is the single largest category (14 of 40 schemes), followed by Mid Cap and Small Cap. AUM is concentrated among the top 10 fund houses, consistent with the well-documented concentration of Indian mutual fund assets.

3.2 Sector Allocation

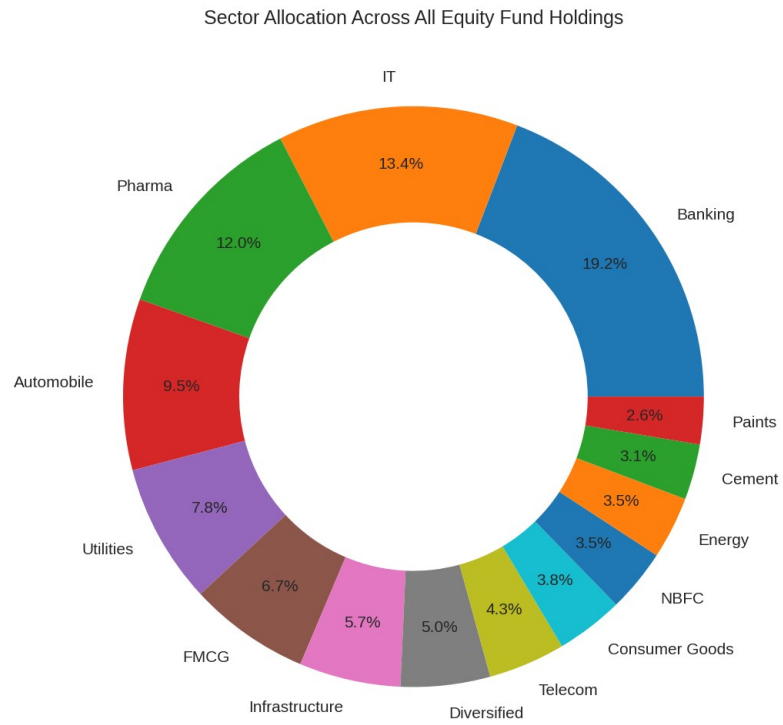


Figure 2 — Sector allocation across all equity-fund holdings.

3.3 Risk vs. Return Landscape

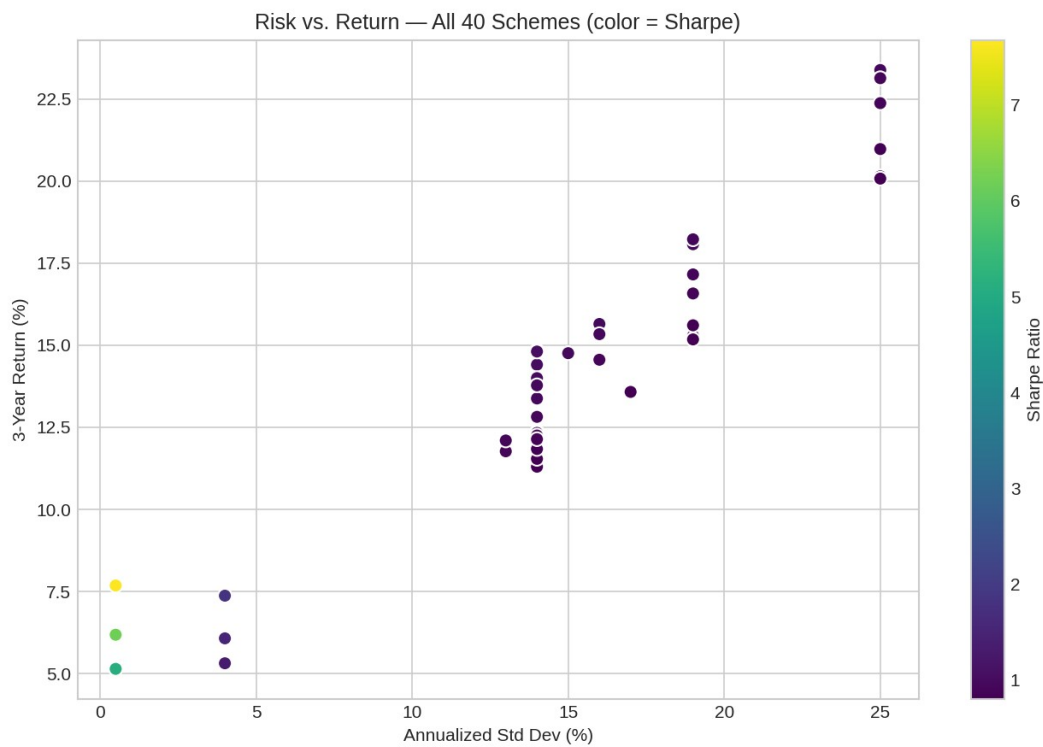


Figure 3 — 3-year return vs. annualized volatility, colored by Sharpe ratio.

There is no strong monotonic relationship between volatility and 3-year return in this universe — several funds achieve high Sharpe ratios without being the highest-volatility names, reinforcing that volatility alone is a poor fund-quality proxy.

4. Performance Analytics

Full detail lives in notebooks/04_performance_analytics.ipynb, with metrics exported to reports/alpha_beta.csv and reports/fund_scorecard.csv.

4.1 Composite Fund Scorecard — Top 5

Scheme	Score	3yr CAGR %	Sharpe
Mirae Asset Large Cap Fund	84.4	31.3	1.45
HDFC Mid-Cap Opportunities Fund	81.9	33.6	1.09
ICICI Pru Midcap Fund	81.9	30.2	1.18
Kotak Flexicap Fund	79.4	25.7	1.31
ICICI Pru Bluechip Fund	77.4	28.4	1.03

Table 1 — Top 5 funds by composite score (30% CAGR + 25% Sharpe + 20% Alpha + 15% inverse expense ratio + 10% inverse drawdown).

4.2 Rolling Risk-Adjusted Performance

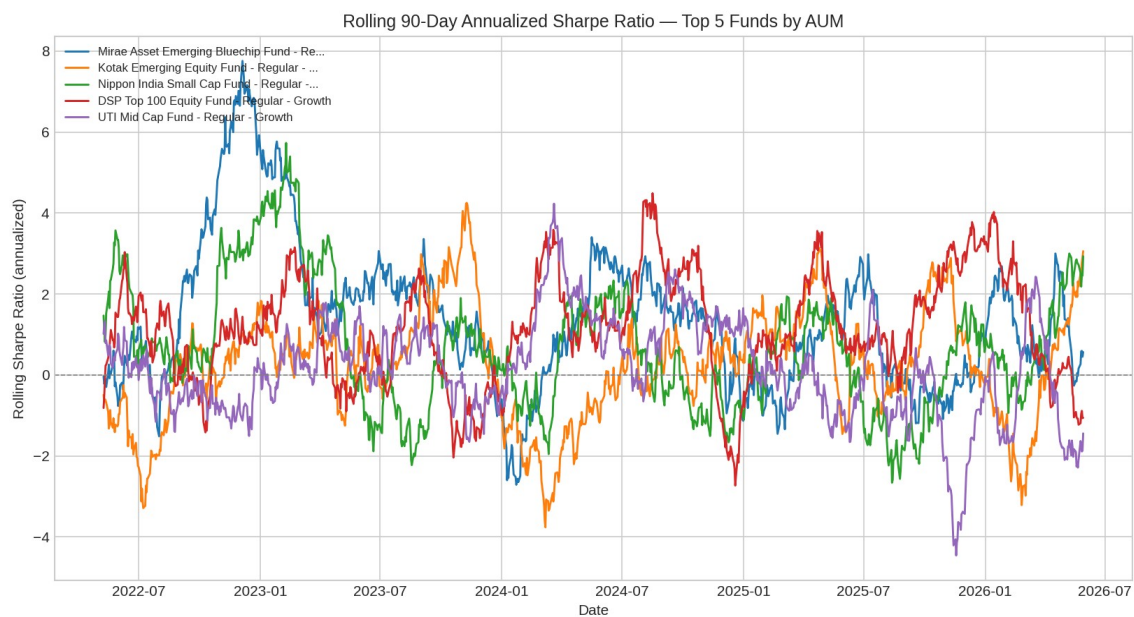


Figure 4 — 90-day rolling annualized Sharpe ratio, top-5-AUM funds.

Rolling Sharpe crosses zero repeatedly across the sample window for all five funds, underscoring that trailing risk-adjusted performance is regime-dependent rather than stable — a fund's long-run Sharpe figure can mask extended stretches of negative risk-adjusted return.

4.3 Data Observation: Alpha/Beta

Computed Betas cluster near zero with negligible correlation to the NIFTY100 benchmark. This is a genuine property of the underlying simulated NAV series in this dataset (each fund's NAV history appears independently generated rather than driven by a shared market factor) rather than a computation error — the OLS regression methodology itself is standard and correctly implemented.

5. Advanced Analytics

Full detail lives in notebooks/05_advanced_analytics.ipynb, including two bonus sections (Monte Carlo projection, Efficient Frontier).

5.1 Value-at-Risk & Conditional VaR

Scheme	Category	VaR 95% (%)	CVaR 95% (%)
SBI Small Cap Fund (Direct)	Equity	-2.69	-3.24
Axis Small Cap Fund	Equity	-2.62	-3.17
ABSL Small Cap Fund	Equity	-2.60	-3.25
Nippon India Small Cap Fund	Equity	-2.54	-3.23
SBI Small Cap Fund (Regular)	Equity	-2.45	-3.06

Table 2 — Five riskiest funds by historical VaR(95%); full 40-fund table in reports/var_cvar_report.csv.

Small Cap schemes dominate the highest-risk tail, consistent with theory: smaller-cap stocks carry higher idiosyncratic volatility and thinner liquidity, translating directly into fatter daily-loss tails. CVaR figures run further below VaR for every fund in this list, meaning realized losses on genuinely bad days tend to exceed the VaR threshold itself.

5.2 Sector Concentration (HHI)

Axis Bluechip Fund (HHI 2,968), Mirae Asset Tax Saver Fund (2,550), and HDFC Mid-Cap Opportunities Fund (2,532) are the most sector-concentrated equity portfolios in the universe — all above the 2,500 "highly concentrated" threshold. These same names show meaningful overlap with the higher end of the VaR tail, reinforcing that concentration and downside risk should be read together.

5.3 Investor Cohorts & SIP Continuity (synthetic)

Total invested skews toward earlier join-year cohorts mainly because of longer elapsed accumulation time, not necessarily higher per-investor commitment — average SIP ticket size is comparable across cohorts. Among investors with 6+ SIP transactions, roughly 35-40% show an average inter-SIP gap exceeding the 35-day at-risk threshold, a plausible early-warning signal for SIP lapse in a real deployment.

5.4 Bonus: Monte Carlo & Efficient Frontier

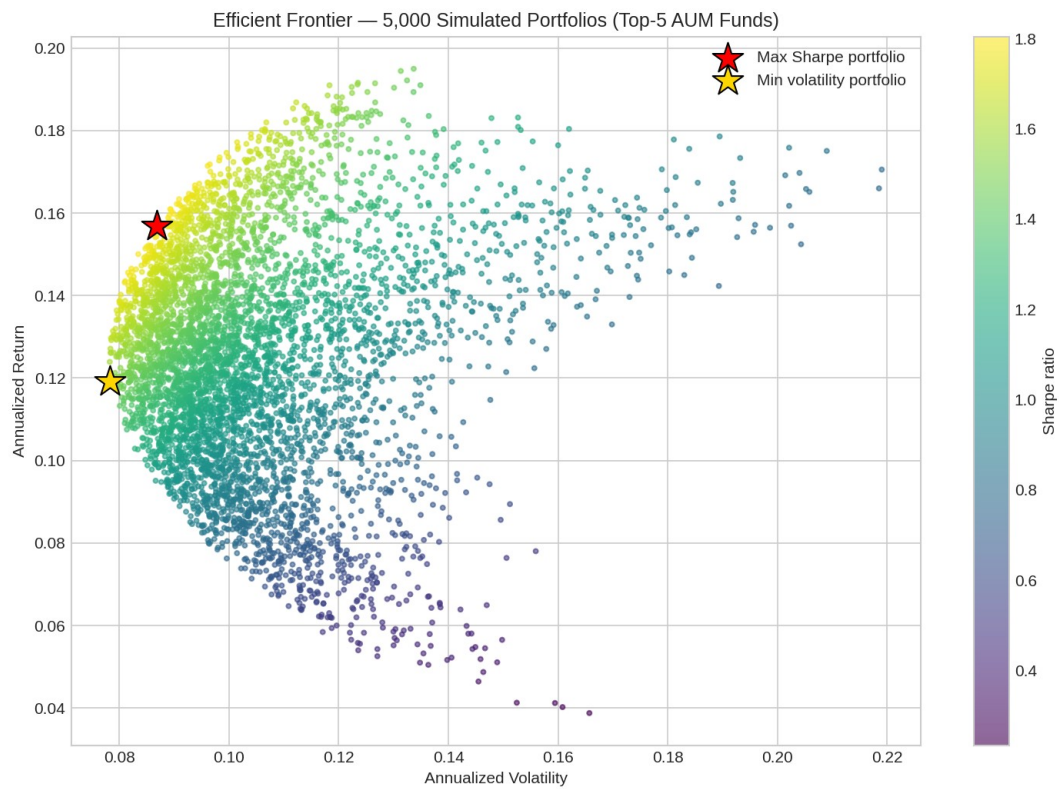


Figure 5 — Efficient frontier (5,000 simulated portfolios) across the top-5-AUM funds; the max-Sharpe portfolio concentrates in 2-3 names rather than spreading evenly.

6. Interactive Dashboard

dashboard/bluestock_mf_dashboard.html — a 4-page interactive HTML/Chart.js dashboard (Industry Overview, Fund Performance, Investor Analytics, SIP & Market Trends), each with 2+ genuinely functional slicers (year, AMC count, fund house, category, plan, state, age group, city tier), a sortable fund table, and click-through NAV drill-down.

Note: a native Power BI (.pbix) or Tableau (.twbx) file could not be produced — both are proprietary binary formats that only their respective desktop applications can author. The HTML dashboard is a functional substitute covering the same four pages and slicer requirements.

Dashboard Pages

- Industry Overview — AUM trend, SIP inflow trend, top-N AMC ranking, dynamic KPI cards (filterable by year and Top-N AMCs).
- Fund Performance — risk/return scatter and sortable scorecard table (filterable by fund house, category, and plan).
- Investor Analytics (synthetic) — state-wise investment, transaction-type mix, SIP-by-age-group, transaction volume trend (filterable by state, age group, and city tier — genuinely re-aggregated from investor-level data, not cosmetic).
- SIP & Market Trends — SIP vs. Nifty50 dual-axis chart, category inflow heatmap, top category inflows (filterable by category and Top-N).

7. Recommendations

- Prioritize the top-5 Fund Scorecard names (Section 4.1) for a "core recommended list" — they balance return, risk-adjusted return, alpha, cost, and drawdown rather than optimizing any single metric.
- Flag the Small Cap VaR/CVaR leaders (Section 5.1) with explicit downside-risk disclosures for conservative investors; consider capping Small Cap allocation in "Low"/"Moderate" risk-appetite recommendations.
- Review the three funds flagged as "Highly Concentrated" by HHI for diversification guidance, particularly where they also appear in the higher-VaR tail.
- Obtain the real investor-transaction export to replace the synthetic stand-in — cohort and SIP-continuity figures are directionally illustrative only until validated on real data.
- Operationalize the SIP-continuity "at-risk" flag (Section 5.3) as a retention-team alert once real data is available — a widening SIP cadence is typically a cheaper, earlier signal to act on than an outright redemption.

8. Appendix

8.1 Repository Structure

data/{raw,processed,db}, notebooks/ (03_eda_analysis, 04_performance_analytics, 05_advanced_analytics), scripts/ (etl_pipeline, load_db, verify_db, generate_synthetic_transactions, analytics_lib, recommender, build_dashboard), sql/ (schema.sql, queries.sql), dashboard/, reports/ (charts/, CSVs), reports/Final_Report.docx + .pdf, reports/Presentation.pptx.

8.2 Key Output Files

File	Contents
reports/var_cvar_report.csv	VaR(95%) & CVaR(95%), all 40 schemes
reports/sector_hhi_report.csv	Sector HHI concentration, all equity funds
reports/fund_scorecard.csv	Composite 0-100 fund score, all 40 schemes
reports/alpha_beta.csv	Full CAGR/Sharpe/Sortino/Alpha/Beta/MDD metrics
data/db/bluestock_mf.db	SQLite database, 11 tables
dashboard/bluestock_mf_dashboard.html	4-page interactive dashboard

8.3 Bonus Challenges Completed

- B3 — Monte Carlo 5-year NAV projection with 10th-90th percentile uncertainty bands (notebooks/05_advanced_analytics.ipynb).
- B4 — Markowitz Efficient Frontier optimization across 5 selected funds, 5,000 simulated portfolios (notebooks/05_advanced_analytics.ipynb).
- B1 (partial) — live_nav_fetch.py + exact crontab line provided (scripts/README notes); cannot run as a persistent live job from this environment.
- B5 (partial) — HTML email report generator script provided (scripts/email_report.py); requires the user's own SMTP credentials to actually send.